

Physiology-Driven Empathic Large Language Models (EmLLMs) for Mental Health Support

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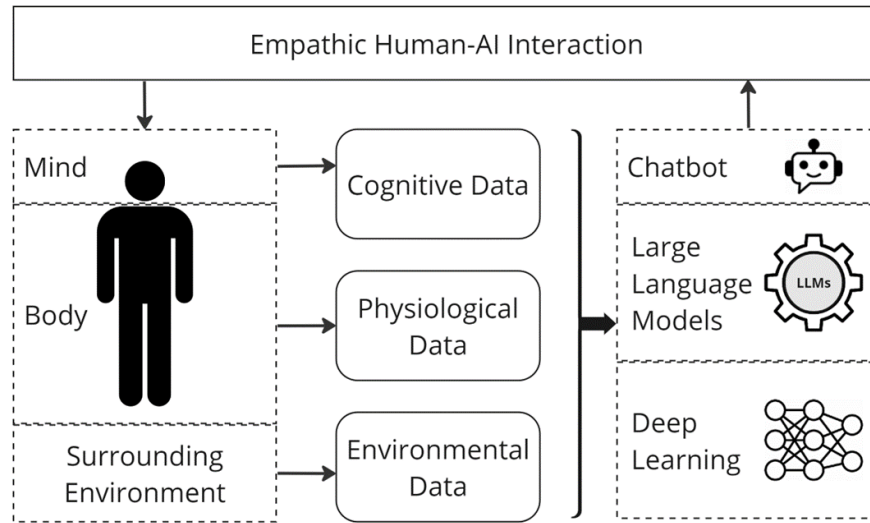


Figure 1: Overview of the proposed approach for automated mental health monitoring and support.

ABSTRACT

Wearable devices show promise in monitoring and managing mental health, but gaps exist in accurately predicting users' mental states and cognitively engaging with users to provide mental health support with wearable data. In this proposal, I present the concept of physiology-driven Empathic Large Language Models (EmLLMs) for mental health support. EmLLMs monitor users and their surrounding environment using wearable devices to predict their mental and emotional states and interact with them based on these states. I present the application of this approach for monitoring and managing excess stress in the workplace. To improve the accuracy of stress prediction, I developed a novel Science-Guided Machine Learning (SGML) model that automatically extracts features from raw wearable data. To engage with users cognitively, I developed an (EmLLM) chatbot that provides psychotherapy based on predicted user stress. I present the SGML model's preliminary findings and results from a pilot user study that evaluates the EmLLM chatbot.

CCS CONCEPTS

• **Human-centered computing** → **User models**; **Interaction paradigms**; **Interaction techniques**; **Interactive systems and tools**; *Ubiquitous and mobile computing*; • **Computing methodologies** → **Machine learning**; **Artificial intelligence**.

KEYWORDS

Wearable Devices, Physiological Data, Deep Learning, Large Language Models (LLMs)

ACM Reference Format:

Poorvesh Dongre. 2024. Physiology-Driven Empathic Large Language Models (EmLLMs) for Mental Health Support. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3651132>

1 ABOUT ME

I am a third-year Ph.D. student in the Department of Computer Science (CS) at Virginia Tech (VT). My supervisor is Dr. Denis Gracanin, Associate Professor in CS at VT. My area of specialization is Human-Computer Interaction (HCI). The anticipated date of my graduation is May 2025. I have never attended a Doctoral Consortium before.

2 INTRODUCTION

In recent times, various factors have contributed to increasing awareness about mental health and wellness. One of these factors is excessive mental stress in the workplace, which significantly affects individuals' health and can also cause other problems, such as anxiety and depression. The 2023 Work in America Survey reported nearly 77% of employees experiencing work-related stress, with 57% indicating negative impacts of occupational stress, such as emotional exhaustion, lack of motivation, and lower productivity [2]. Therefore, it is important to track mental stress in the workplace and provide necessary interventions to reduce its negative impact.

Wearable devices are emerging as promising platforms for tracking users' mental health. It collects users' physiological data, such as Electrodermal Activity (EDA), Blood Volume Pressure (BVP), and Skin Temperature (ST), which can be used to predict their psychological states, such as stress and emotions. The process of uncovering patterns and relationships between users' physiological data and their psychological states is called Psycho-physiological Modeling (PsPM). Data from wearable devices can also be used to provide biofeedback to the users. Biofeedback is a therapeutic technique that provides users with information about their physiological functions by giving them visual, auditory, or haptic cues.

In this research, I focus on the following challenges pertaining to PsPM and biofeedback: (1) Methods that typically infer psychological states from selected physiological features instead of the entire raw physiological data may suffer from information loss [3]. (2) Moreover, using such methods requires domain expertise to extract relevant features from physiological data [20], and there is no consensus on the best physiological features; (3) Biofeedback alone is not always effective in providing therapeutic support, and it should be accompanied by techniques that can cognitively engage with users [6].

To address these challenges, I present the concept of physiology-driven Empathic Large Language Models (EmLLMs) for empathic human-Artificial Intelligence (AI) interaction (Figure 1). It involves monitoring users and their surroundings using wearable devices and using advanced Machine Learning (ML) techniques to automatically extract features from raw physiological data to predict their psychological states. It further involves customizing pre-trained LLMs to develop chatbots that can cognitively engage with the users based on their predicted psychological states. I present the application of this approach to monitoring and managing the mental stress of individual users in their workplaces.

3 RELATED WORK

In psychophysiology, stress is defined as a complex, multi-dimensional response of the human mind and body to various stressors. Stressors can be external factors, such as poor environmental conditions, or internal factors, such as emotions, that trigger stress responses. A stress response typically involves physiological, psychological, and behavioral changes, such as increased heart rate, negative emotions, and heightened alertness. This response can help individuals respond to challenges in the short term, but chronic stress can negatively impact health.

3.1 Psycho-physiological Modeling of Stress

Wearable sensors are emerging as promising platforms for personalized and accessible stress monitoring. Prior works on using wearable sensors for monitoring stress use various physiological signals, such as BVP, EDA, ST, and Electroencephalogram (EEG), and extract time-domain, frequency-domain, and time-frequency features including mean, power-spectral density, and wavelet coefficients, from the raw physiological data [15, 19, 32, 34]. Supervised ML techniques, such as Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM), are applied to selected features for predicting user stress [16, 34]. Unsupervised ML techniques such as fuzzy logic and Principal Component Analysis (PCA) are also used for data pre-processing and feature extraction. However, there is no consensus on the best physiological features, and researchers typically use the features that give them the best accuracy.

Deep Learning (DL) provides unique opportunities to incorporate the feature engineering process in the modeling task and remove the dependence on manually designed features [20]. DL networks, such as Convolution Neural Networks (CNNs), automatically extract useful time-varying features from raw physiological data [39]. CNNs can be combined with recurrent neural networks to develop hybrid networks for time-series classification of human activities and states using wearable data [11], [21], [38]. DL networks also provide the flexibility of taking multi-modal sensor data as separate inputs and merging them within the network [40], thus eliminating the need to reshape irregularly sampled physiological data. Moreover, DL networks allow Multi-Task Learning (MTL) of related tasks in one network, resulting in improved learning efficiency and prediction accuracy [7]. Lastly, DL networks can also enable the creation of relevant features for sensor data by following a science-guided approach [25]. However, there is little to no research on using the discussed DL techniques for PsPM of stress and other psychological states.

3.2 Automated Solutions for Mental Health Support

Data from wearable devices is also used to provide biofeedback to users to regulate stress and manage other mental health concerns. Biofeedback provides users with real-time visual, auditory, or haptic information about their physiological processes, which allows them to control these processes. Pallavicini et al. [29] showed users visuals of various artifacts moderated through Heart Rate (HR) in Virtual Reality (VR) to treat anxiety. Harris et al. [17] proposed an audio biofeedback technique that encourages slow breathing by adjusting the quality of music in proportion to the user's respiration rate to reduce stress. AmbienBeat is a wearable device that provides biofeedback via tactile stimulus based on users' HR for stress regulation [8]. There is no consensus on the best biofeedback modality for regulating stress and managing other mental health concerns, but a common limitation of these modalities is that they fail to engage users cognitively.

Psychotherapy techniques, such as Cognitive Behavior Therapy (CBT), involve cognitive engagement with the user and help them identify distortions in their thought processes. With the advancement of technology, there are now Artificial Intelligence (AI) conversational agents or chatbots that provide psychotherapy to

users. Woebot, a fully automated conversational agent, effectively delivered accessible mental health support to college students [14]. Users must actively engage with the chatbot and inform it about their mental state, but it's not always possible for users to keep track of their mental state throughout the day; wearable devices can help track them automatically. BioBase combines elements of wearable tracking and psychotherapy, but the wearable device is used for passive sleep, HR, and physical activity tracking, and a separate mobile phone app is used to assess users' moods [30]. Another limitation of existing AI chatbots is their fixed conversational style of interacting with users, which can feel generic and robotic [14].

The emergence of Large Language Models (LLMs) offers new and unique opportunities to provide personalized solutions for stress regulation and mental health support. Pre-trained LLMs such as GPT-3 can perform various natural language processing and generation tasks. However, chat models such as ChatGPT is an application of the GPT-3 model and require some customization for specialized applications. Prompt engineering is a technique for customizing the behavior of LLMs by giving them a set of instructions to perform better on particular tasks. With prompt engineering, LLMs can be instructed to have the personality of a psychologist and can be further instructed to respond to users in a specific way [28, 37]. Model fine-tuning refers to the process of updating the parameters of a pretrained LLM on a task-specific dataset. To customize an LLM for mental health support, it can be fine-tuned on various mental health datasets containing question-answer pairs related to psychological counseling [26]. LLMs can also be trained from scratch on large-scale mental health data [23], but this process requires more time and resources. Using LLMs for mental health applications is a relatively new area of research and needs more exploration.

4 PRELIMINARY, COMPLETED, AND ONGOING WORK

In this section, I first describe my preliminary works, which reveal the concern about mental stress among university students and the impact of stressors on human behavior in university buildings. The second part discusses the development of advanced ML models that automatically extract features from raw physiological data to predict user stress. Lastly, I discuss the development and usability testing of a chatbot based on the EmLLM concept for psychotherapy that dynamically interacts with users based on their predicted stress levels.

4.1 Preliminary Work

4.1.1 Learning Experiences in University Buildings. The building environment greatly impacts the health and productivity of occupants. To evaluate the impact of the building environment on students' learning experiences, I conducted focus groups and semi-structured interviews among 12 undergraduate and 7 graduate students who spent more than 10 hours per week in a university building at VT [31]. The results of this qualitative study revealed that "stress" was a major concern for graduate students because of their studies. Poor Indoor Environment Quality (IEQ) conditions, including noise, bad lighting, and thermal comfort, added to their existing mental stress. Among all IEQ conditions, students reported thermal comfort as

a significant occupant-centric problem. Thermal comfort is a psychological state that indicates an individual's perceived satisfaction with their surrounding environment [35].

4.1.2 Thermal Comfort Behavior in University Buildings. Later, I investigated the impact of thermal comfort on occupant behavior and the impact of this behavior on the indoor temperature and thermal comfort of other occupants in the university building [10]. I used the human-building interactions dataset by Langevin et al. [24] to develop explainable ML models to predict if occupants will turn the thermostat up or down at a given indoor temperature. These models were used to simulate the impact of thermostat usage behavior on indoor temperature and overall thermal comfort levels. Simulation results show that the final indoor temperature always drifted to a cooler temperature, indicating that the behavior of occupants who prefer to have cooler conditions is dominant over that of other occupants.

4.1.3 Motivation Moving Forward. The concerns about mental stress among graduate students found in my preliminary research motivated me to investigate technological solutions to monitor mental stress and provide mental health support. Other studies also highlight the concerns of mental health among graduate students [33]. However, despite this dire situation, graduate students struggle to seek support due to fears of harming their academic careers, financial insecurity, and lack of time [36]. Therefore, I focused on solutions that are accessible and personalized, and as a result, I navigated towards using wearable devices and AI chatbots in my research.

4.2 Deep Learning for Psycho-Physiological Modeling of Stress

My preliminary work found that students' stress was primarily a consequence of their academic pursuits, which were exacerbated by thermal comfort. Thermal comfort is a psychological state that can be predicted with physiological data [22]. Therefore, in this section, based on the advantages of DL discussed in Section 3.1, I explore the following research goals:

Goal 1: Develop an end-to-end DL model to predict stress and thermal comfort from multi-modal wearable data

Goal 2: Train and test the model on various scientific datasets that collect psycho-physiological data

Goal 3: Modify and update the proposed DL model to give state-of-the-art accuracy

To achieve these goals, I first developed an end-to-end multi-channel 1D CNN model for stress prediction that inputs the raw physiological data individually through separate channels. It then applies a sequence of 1D convolutions, pooling (1D max-pooling), and flattening on each channel before concatenating and passing it over to fully connected layers for classification. I trained this model on the Wearable Stress and Affect Detection (WESAD) dataset [34] for binary stress classification. This model achieved a state-of-the-art accuracy of 85.1% and 89.0 F1 score.

Next, I developed an end-to-end multichannel hybrid DL model using 1D CNN, Long Short-Term Memory (LSTM), Attention, and fully connected dense layers with soft parameter sharing for MTL of thermal preference and thermal sensation. I trained and tested this

model on the raw dataset from a study by Liu et al. [27] to predict thermal preference (cooler, no change, and warmer) and thermal sensation (cold, neutral, and warm) for individual participants. This model achieved an *average accuracy of 66% and 56%* to predict thermal preference and sensation, respectively. However, the state-of-the-art ML (RF) model trained on hand-crafted features to predict thermal preference achieved an *average accuracy of 72%*. This could be because DL models often require more data than traditional ML models.

Another limitation of the proposed hybrid model is that it can only extract time-varying features from the raw physiological data. Traditional time-domain analysis may not capture the dynamic nature of physiological data, and time-frequency analysis is needed to uncover patterns in physiological phenomena. To overcome this limitation, I am developing a DL model that uses 1D CNNs to mimic the physics of a high-pass and low-pass filter in a Discrete Wavelet Transform (DWT). The preliminary performance of this model on the WESAD dataset to predict stress and emotions shows promising results.

4.3 Physiology-Driven Empathic Large Language Model (EmLLM) Prototype

In this section, I discuss the development and usability testing of a physiology-driven EmLLM chatbot that monitors user stress throughout the day using their wearable data and interacts with them at the end of the day based on their stress levels. LLMs are designed to understand and generate human-like text and can mimic empathy but don't actually understand and feel emotions. Moreover, LLMs aren't inherently designed to understand or analyze wearable data. Therefore, to develop a physiology-driven EmLLM chatbot for stress monitoring and management, I explore the following research goals:

- Goal 1:** Customize a pre-trained LLM for stress-related psychotherapy
- Goal 2:** Integrate the wearable data for monitoring stress with the customized LLM to develop the EmLLM chatbot
- Goal 3:** Evaluate the effectiveness of the EmLLM chatbot in providing mental health support to users for stress management

To that end, I use the 1D CNN model for binary stress classification discussed in Section 4.2 to predict if a participant is feeling "stressed" or "not stressed". I fine-tune the Falcon-7B LLM [1] using the Quantized Low-Rank Adaption (QLoRA) technique [9] on the mental health conversational dataset [5]. This dataset has a rich collection of conversational pairs, including patient questions and corresponding answers provided by mental health professionals. Next, the fine-tuned LLM is prompt-engineered to take as input the predicted user stress, act like a trained psychologist, follow the principles of CBT, and provide a reasonable explanation whenever it cannot answer a question. Lastly, I develop a web interface for the EmLLM chatbot for user interaction.

I conducted a pilot user study to evaluate the EmLLM chatbot's ability to accurately predict user stress, provide human-like responses, and assess the therapeutic support provided to the user. Participants (8 Ph.D. students, 5 men, and 3 women between ages 23 and 37) were passively monitored throughout their workday

using the Empatica EmbracePlus smartwatch [13]. They completed the Short Stress State Questionnaire (SSSQ) [18] before starting and after finishing their workday. At the end of the day, participants were asked to interact with the EmLLM chatbot for at least 15 minutes. The chatbot greets the participants, informs them about their predicted stress, and asks if they want to discuss their day with it. After completing the session, the participants were asked to complete the Godspeed questionnaire [4] and the Session Rating Scale (SRS) [12]. Preliminary results indicate that the EmLLM chatbot can accurately predict stress, provide a human-like response, and develop a good therapeutic alliance with most users.

5 DISCUSSION, CONCLUSION, AND FUTURE WORKS

Currently, the entire world is experiencing a shortage of trained mental health professionals. Moreover, there also exist various social, cultural, and economic barriers to seeking mental health support. AI chatbots offer a promising solution to overcome these challenges, and my research aims to advance such chatbots to provide accessible and personalized mental health support.

I propose the concept of EmLLMs that can interpret users' psychological states from their wearable data by integrating LLMs with advanced ML models. I explore the development of various end-to-end DL models to address the challenges associated with manual feature engineering in PsPM. I developed a novel SGML model that automatically extracts time-frequency features from raw physiological data to predict psychological states. I also discuss the design and evaluation of an EmLLM chatbot for stress-related psychotherapy. EmLLMs can also be used for other applications, such as monitoring elderly health, training for special jobs, and creating virtual companions.

For future work, I want to include other psychological states related to stress in the workplace in the prediction model. This will give the EmLLMs more information about users and their surrounding environment and allow them to engage in more meticulous discussions to address their stress levels. I also intend to explore more reliable ways of customizing the LLMs for stress-related psychotherapy. Lastly, I want to conduct a robust user study with more participants and extensive protocols on the EmLLM chatbot for stress monitoring and management.

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